THE N-2 PROPERTY ASCENDS ALONG A QUASI-FINITE EXTENSION OF NOETHERIAN DOMAINS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Quasi-finiteness gives finiteness of the induced fraction field extension. Used at the start of the counted proof (lines 45425-45429). Provides the finite field extension that the N-2 property of the base is applied to.
- The integral closure as a finite module over an N-2 base. Used in the middle of the counted proof (lines 45430-45446). Produces a finite intermediate ring whose integral closure over the target is the object whose finiteness has to be shown.

Source reference. Original lemma 032H: sources/stacks-algebra/pinned-original-source.tex, source lines 45402–45402. Navigation pointer for \ref{lemma-Japanese-local} used inside the counted span (line 45438); the argument that consumes it is retained in the span.

Source reference. Original lemma 0308: sources/stacks-algebra/pinned-original-source.tex, source lines 7940–7940. Navigation pointer for \ref{lemma-integral-closure-transitive} used inside the counted span (line 45446); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02JM: sources/stacks-algebra/pinned-original-source.tex, source lines 7927–7927. Navigation pointer for \ref{lemma-integral-permanence} used inside the counted span (line 45448); the argument that consumes it is retained in the span.

Source reference. Original lemma 00PK: sources/stacks-algebra/pinned-original-source.tex, source lines 30413–30413. Navigation pointer for \ref{lemma-isolated-point-fibre} used inside the counted span (line 45427); the argument that consumes it is retained in the span.

Source reference. Original lemma 032G: sources/stacks-algebra/pinned-original-source.tex, source lines 45390–45390. Navigation pointer for \ref{lemma-localize-N} used inside the counted span (line 45438); the argument that consumes it is retained in the span.

Source reference. Original lemma 00QB: sources/stacks-algebra/pinned-original-source.tex, source lines 31123–31123. Navigation pointer for \ref{lemma-quasi-finite-open-integral-closure} used inside the counted span (line 45434); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a domain. Let $R \subset S$ be a quasi-finite extension of domains (for example finite). Assume R is N-2 and Noetherian. Then S is N-2.*

Proof. Let L/K be the induced extension of fraction fields. Note that this is a finite field extension (for example by Lemma 00PK (2) applied to the fibre $S \otimes_R K$, and the definition of a quasi-finite ring map). Let S' be the integral closure of R in S. Then S' is contained in the integral closure of R in L which is finite over R by assumption. As R is Noetherian this implies S' is finite over R. By Lemma 00QB there exist elements $g_1, \ldots, g_n \in S'$ such that $S'_{g_i} \cong S_{g_i}$ and such that $g_1, \ldots, g_n$ generate the unit ideal in S. Hence it suffices to show that S' is N-2 by Lemmas 032G and 032H. Thus we have reduced to the case where S is finite over R.

Assume $R \subset S$ with hypotheses as in the lemma and moreover that S is finite over R. Let M be a finite field extension of the fraction field of S. Then M is also a finite field extension of K and we conclude that the integral closure T of R in M is finite over R. By Lemma 0308 we see that T is also the integral closure of S in M and we win by Lemma 02JM. $\square$